

Addendum: Reproducibility Clarification — Coordinate Dependence of k

This addendum clarifies an implementation detail required for exact reproduction of the collapse fits reported in “Consciousness Collapse: Loss of Consciousness as Impedance Catastrophe.”

Model:

$$R(\eta) = 1 / (1 + \exp(k (\eta - \eta_{\text{mid}})))$$

Clarification (important):

The steepness parameter k is not invariant under linear rescaling of η . If η is normalized (e.g., z-scored) within each perturbation domain prior to pooling, then η is mapped by

$$\eta' = a \eta + b$$

and the fitted steepness rescales as

$$k' = k / a.$$

Practical consequence:

- A pooled fit using raw η can yield k values numerically different from a pooled fit using domain-normalized η , even when both fits describe the same collapse manifold and produce comparable R^2 / residual structure.
- For cross-domain universality tests, report k in the same η coordinate used for fitting, and state the η normalization explicitly (recommended: domain-normalized η with 0 = baseline).

Reproducibility note:

The accompanying GitHub package includes a reference notebook that performs domain-normalization of η prior to fitting and bootstraps k , ensuring deterministic reproduction given the same CSV.

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